Bill of Materials (BOM)

i-meic-pico_V6 Board

DOS PC with an 8088 single-board computer (SBC)

Project: i-meic – single-board DOS PC
As of: Juli 2026 / July 2026
https://github.com/DerHobbyEntwickler/i-meic

Processing core: CPU Intel 80C88 + 2× SRAM AS6C4008 (512 KB each, 1 MB total max.)

Line items: **22** · Total parts: **56**

Part	Qty	Value	Remarks	Note	Datasheet
C1, C14	2	470uF/10V or 1000uF/10V	Electrolytic capacitor, radial, D8.0 mm, P3.50 mm	**H01**	Datasheet
C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13, C18	13	1uF/50V or 10uF/50V	Monolithic ceramic capacitor, pitch 2.5 mm to 5 mm	**H02**	Datasheet
D1	1	LED green, UART	Light-emitting diode, 3 mm or 5 mm	**H03**	Datasheet
D2, D17	2	LED white, SRAM	Light-emitting diode, 3 mm or 5 mm	—	Datasheet
D3, D4, D5, D6	4	LED blue, ITP3	Light-emitting diode, 3 mm or 5 mm	—	Datasheet
D7, D8, D9, D10, D11	5	1N4001	Diode 1N4001	**H04**	Datasheet
J10	1	Connector for serial adapter	Pin header 2.54 mm, single row, 4 pins	**H05**	Datasheet
JP1	1	Jumper connector, 3 pins	Pin header 2.54 mm, single row, 3 pins	—	Datasheet
JP2	1	Jumper connector, 2 pins	Pin header 2.54 mm, single row, 2 pins	—	Datasheet
J_DS0, J_DS1, J_DS2, J_DS3	4	Connector for ITP3 cable, dual row	Box header, 10-pin, straight, dual row	**H06**	Datasheet
R1	1	1K	Resistor, carbon film, 1.0 kOhm, 0207, 250 mW, 5 %	**H07**	Datasheet
R2, R15	2	6.8K	Resistor, carbon film, 6.8 kOhm, 0207, 250 mW, 5 %	—	—
R4, R5, R6, R7	4	2.2K	Resistor, carbon film, 2.2 kOhm, 0207, 250 mW, 5 %	—	—
R8, R9, R11	3	10K	Resistor, carbon film, 10 kOhm, 0207, 250 mW, 5 %	**H08**	—
SW2	1	Reset button	Tact switch, PCB mount, 1 NO contact, 12 x 12 x 4.3 mm	—	Datasheet
U1	1	IC 74HC00	Quad 2-input NAND gate, 2 … 6 V, DIL-14	**H09**	Datasheet
U2	1	IC 74HC02	NOR gate, 2-input, 2 … 6 V, DIL-14	—	Datasheet
U3, U10, U13	3	8-channel bidirectional logic level shifter (3.3V ↔ 5V)	8-channel bidirectional logic level shifter module (3.3V ↔ 5V)	—	Datasheet
U4	1	IC CPU 80C88	CPU Intel 80C88 or compatible	**H10**	Datasheet
U5, U6	2	IC 74HC373	Latch, 3-state, 2 … 6 V, DIL-20	—	Datasheet
U7, U8	2	IC SRAM AS6C4008-55PCN	SRAM, 4 Mb (512 K x 8), 4.5 … 5.5 V, PDIP-32	**H11**	Datasheet 1 · Datasheet 2

Part	Qty	Value	Remarks	Note	Datasheet
U12	1	RP2040 Pro Micro / Raspberry Pi Pico	RP2040 Pro Micro / Raspberry Pi Pico	**H12**	RP2040 Pro Micro Raspberry Pi Pico development board (source: AliExpress)

Blue underlined entries in the "Datasheet" column are clickable links.

Notes on the bill of materials

H01 — 470uF/10V or 1000uF/10V

2× electrolytic capacitors used as decoupling capacitors. Values are not critical.

H02 — Ceramic capacitor, 1uF or 10uF

13× decoupling capacitors, values are not critical. For decades a 100 nF decoupling capacitor was used for every IC. Today, ceramic capacitors from 1 uF upward are very cheap. Recommendation: 10 uF is even better.

H03 / H07 — LEDs and their series resistors (carbon film, 1K / 6.8K / 2.2K)

In principle, the resistors listed here are the series (dropping) resistors for the various LEDs. They are classically calculated by subtracting the LED forward voltage from the supply voltage. The resulting voltage, together with a chosen LED current (e.g. 5 mA), is used to calculate the resistor value. In practice, LED quality and typical on-time also matter a great deal.

The two white LEDs (R2, R15) are very bright and need only very little current for normal operation, so a series resistor of 6.8K has proven sufficient here. R1 with 1K as the series resistor for the green UART LED, by contrast, is often somewhat dim – especially since the supply voltage here is only 3.6 V – or a super-bright green LED is used. The 2.2K series resistor for the four blue ITP3 LEDs (R4, R5, R6, R7) has proven itself in practice.

H04 — Diode 1N4001

The value and type of the diode are entirely uncritical.

H05 — Pin header

The various pin headers (with 2, 3 and 4 pins) can be cut from one long 40-pin header. Coloured headers improve orientation but are not strictly necessary.

H06 — Box header (ITP3 cable)

No further text is provided for this note in the source. These are the 10-pin, dual-row box headers (J_DS0…J_DS3) for the ITP3 cables.

H08 — Carbon-film resistors 10K

Pull-up resistors to ensure a constant voltage level. Resistor values are not critical (4.7K … 22K should meet the requirements).

H09 — 74HC00 and 74HC02

Standard 74-series logic. Also relatively uncritical (HC, HCT or LS are possible).

H10 — CPU Intel 80C88 or compatible

The central CPU IC, the 8088, has become quite rare by now and is somewhat more expensive. Where possible, the CMOS variant should be preferred (lower power consumption, runs cooler). The NMOS variants should also be usable (but this has not yet been tested).

One more note: cheaper components are often available on AliExpress. They were frequently used in these prototypes. However, at least in part, these appear to be used goods. That is not a problem in itself. What can happen is that the contacts are more or less oxidised. If sockets are used (which is recommended), contact difficulties may occur during assembly. In that case, please clean the IC pins (with a glass-fibre pen, carefully with fine sandpaper, or carefully with a small file).

H11 — SRAM AS6C4008-55PCN or pin-compatible IC

Components from German suppliers (such as Reichelt Elektronik) were always of high quality and without complaints, but also noticeably more expensive. On AliExpress they can certainly be obtained more cheaply, but may then be of lower quality (see “One more note” under H10).

H12 — RP2040 Pro Micro / Raspberry Pi Pico

The RP2040 Pro Micro board is considerably smaller than the classic Raspberry Pi Pico. For this reason it was used for this board. Both boards contain an RP2040 and are software-compatible. The source for the RP2040 Pro Micro board is AliExpress.